Software Engineering Department

Braude College of Engineering

Final Project – Phase A (61998)

Characterizing Normal Brain Development via EEG Pattern Matching

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By:

NOOR AMORY

WADIEA FARRAN

Advisor:

Dr. SAMAH IDREES GHAZAWI

Link to GitHub:

https://github.com/wadiea1/Eeg-age-patterns.git

Abstract

Understanding how the human brain develops across childhood and adolescence is essential for both neuroscience and clinical research, yet most EEG studies focus on disorders rather than normative developmental patterns. This project will investigate whether identifiable and recurring structural patterns exist in EEG signals across different age groups and how these patterns evolve during brain maturation. Using a dataset of multi-channel EEG recordings from healthy participants, we will preprocess the signals through filtering, normalization, segmentation, and frequency-band extraction to obtain stable representations of the underlying neural activity. We will then analyze these segments using computational pattern-detection techniques inspired by stringology and signal comparison to identify EEG components that consistently appear in specific age groups but not in others. By mapping frequency-specific and structurally preserved patterns across age ranges, this study aims to build an initial reference framework for normal brain development. Such findings may support future efforts in early detection of atypical neural development and will contribute to more data-driven models of brain maturation.

Keywords: Electroencephalography (EEG), Brain Development, Neural Patterns, Signal Normalization, Pattern Detection.

1. Introduction

In this project, we will analyze EEG data collected from healthy children and adolescents across multiple age groups to characterize normal brain development rather than a specific clinical disorder. The participants are divided into age-based cohorts, allowing us to compare how EEG patterns change from childhood into late adolescence.

Recordings were obtained while participants performed a small set of standardized tasks, including resting-state and simple visual or attention-related paradigms designed to elicit robust and reproducible brain responses under comparable conditions across subjects and age groups.

EEG was recorded using a high-density 128-channel system with a uniform sampling rate and common reference electrodes, as specified in the dataset documentation [1]. These multi-channel, time-synchronized recordings form the basis for all subsequent preprocessing, normalization, segmentation, and pattern-detection analyses presented in this work.

1.1 The objective of the project

This project aims to analyze the collected EEG data from healthy children and adolescents in order to identify age-dependent patterns in brain activity and characterize normal brain development. Specifically, we seek to detect recurring wave and frequency structures that are typical for certain age groups and to compare how these patterns change across development.

To achieve this, we will first preprocess the EEG signals using filtering, artifact removal, segmentation into time windows, and frequency-band extraction, followed by appropriate normalization strategies to reduce inter-subject variability. Afterwards, we will apply computational pattern-detection methods inspired by stringology algorithms such as Order Preserving Matching (OPM) and Cartesian Tree Matching (CTM), to locate recurring temporal and spectral structures within and between age groups.

A central goal is to evaluate whether some of these patterns can serve as stable markers of typical brain maturation, and to examine how different normalization strategies influence the robustness and separability of age-group representations. In the long term, the project aims to provide an initial reference framework for normal EEG development that may support future tools for early detection of typical neural trajectories.

1.2 The Problem

Current EEG studies often emphasize classification of disorders or task-specific outcomes, while less attention is given to discovering stable, reusable representations of *normal* developmental patterns across ages. This creates three core challenges:

1. High inter-subject variability in EEG amplitudes and noise characteristics.

2. Difficulty separating developmental effects from differences caused by recording conditions or preprocessing choices.

3. Lack of compact pattern libraries that summarize typical neural structures for each age stage.

As a result, it remains difficult to define computational markers that consistently represent normal maturation trajectories. This project addresses that gap by focusing on structural pattern consistency rather than raw amplitude similarity.

1.3 Review of Existing Solutions

Existing solutions in the literature can be grouped into three directions:

- Normalization-centered pipelines: Studies on the HBN-EEG dataset show that preprocessing and normalization choices strongly influence downstream model behavior, especially in cross-subject settings.

- Cross-task / cross-subject learning frameworks: Foundation-style EEG approaches attempt to generalize across tasks and participants, highlighting both the potential and the challenge of robust representation learning.

- Deep hierarchical models: Neural architectures have been proposed for brain-state prediction and health assessment, demonstrating strong predictive ability but often with limited interpretability of recurring structural motifs.

Compared with these approaches, our method emphasizes interpretable structural matching (OPM/CTM) and motif discovery to produce age-group-specific pattern templates that can be directly inspected and compared.

1.4 Definitions

EEG (Electroencephalography): A non-invasive technique that measures the brain's electrical activity via electrodes on the scalp, capturing voltage fluctuations from neuronal activity. It offers high temporal resolution, making it ideal for studying rapid brain processes across five key frequency bands: Delta, Theta, Alpha, Beta, and Gamma. In this project, it serves as the primary data source for characterizing age-dependent neural development.

OPM (Order-Preserving Matching): A pattern-matching method that focuses on the relative rank-order of signal elements rather than their absolute numerical values [8]. This allows researchers to identify recurring structural waveforms that remain consistent even when amplitudes vary significantly between different subjects. It is particularly effective for detecting functional brain patterns that maintain their shape during maturation despite physiological differences.

CTM (Cartesian Tree Matching): A technique that represents EEG segments as binary trees to capture hierarchical relationships and the nested structure of peaks and valleys. Two sequences are considered a match if they produce identical tree structures [5], allowing for the comparison of global organizational principles independent of signal scale. In this study, it complements OPM by identifying broader structural principles underlying cognitive development.

Motif Discovery: The computational process of identifying short, recurring shapes within EEG data that appear frequently enough to represent "typical" neural templates. Once these motifs are extracted and encoded using OPM or CTM, they form a library of patterns specific to different age groups. These templates then serve as markers to enable faster and more robust matching [9] when analyzing new subjects.

2. Background and Related Work

To understand brain activity patterns across different age groups, we first explore the principles of EEG and its potential to identify age-dependent changes in electrical brain patterns.

2.1 EEG Fundamentals and Frequency Bands

Electroencephalography (EEG) measures the brain's electrical activity through electrodes placed on the scalp, capturing voltage fluctuations from neuronal activity in the cerebral cortex. With high temporal resolution, EEG is ideal for studying rapid brain processes, though its spatial resolution is limited due to overlapping signals from different regions.

This non-invasive technique records brain waves across five primary frequency bands, as illustrated in Fig.1 below, where the x-axis represents time and the y-axis represents amplitude. Each band is associated with specific cognitive states:

Delta (0.5–4 Hz): Deep sleep and restorative functions

Theta (4–8 Hz): Relaxation and meditation, prominent in younger children

Alpha (8–12 Hz): Calm alertness, increasing with age

Beta (12–30 Hz): Focus, concentration, and problem-solving

Gamma (>30 Hz): Complex cognitive tasks like memory and attention

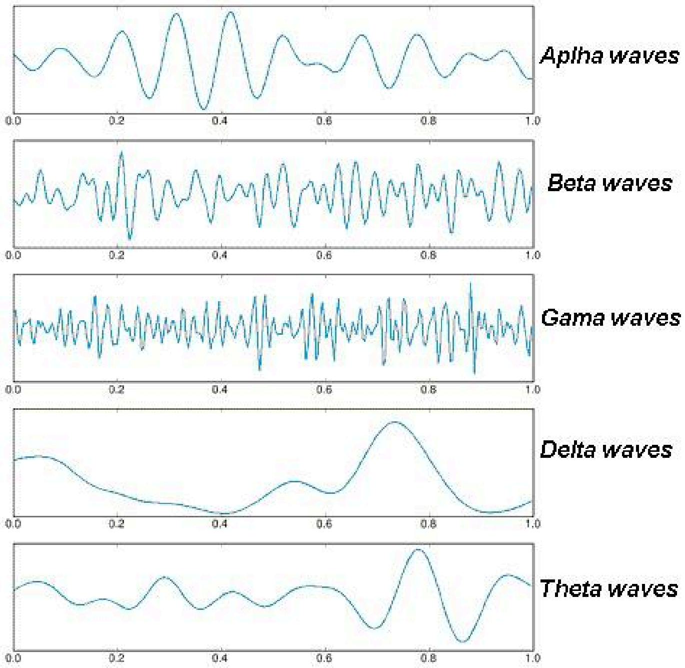


Fig. 1: Visual representation of the five primary EEG frequency bands used for neural pattern analysis. This comparison illustrates the varying amplitudes and frequencies of Delta, Theta, Gamma, Beta, and Alpha X-axis: Time (s); Y-axis: Amplitude (V).

2.2 Pattern matching

Pattern matching is an important technique for identifying recurring structures in EEG signals, which can help reveal specific brain activities or cognitive states across different age groups. In EEG analysis, the goal is to detect patterns that repeat over time and may indicate underlying neural processes, such as attention or relaxation. For this project, we will use pattern matching algorithms to identify these recurring waveforms and structures, allowing us to track how brain activity changes with age. Specifically, we will apply Cartesian

Tree Matching (CTM) and Order-Preserving Matching (OPM) to detect and compare patterns in EEG signals. These methods help us focus on the relative relationships between different signal components, ensuring we accurately capture the structural features of the data. Through this approach, we aim to uncover consistent patterns of brain activity that correlate with normal brain development.

2.2.1 Order-Preserving Matching (OPM) in EEG Analysis

Order-Preserving Matching (OPM) is a pattern matching technique that focuses on the relative ordering of elements rather than their exact values [8]. In EEG signal analysis, this method allows us to identify recurring brainwave patterns that maintain their structural relationships over time, independent of absolute amplitude values. This is particularly valuable when analyzing brain activity data where absolute values may vary significantly across subjects or recordings, but the relative order of signal changes holds important neurological meaning.

The algorithm converts both the pattern and the EEG signal segments into relative ordering representations. For example, a pattern [100, 150, 120] becomes [0, 2, 1] based on relative ranks [6]. The algorithm then slides through the EEG signal, comparing the relative ordering of each window against the pattern's ordering. When the orderings match, a pattern occurrence is detected. This process is performed using the KMP-Order-Matcher algorithm Fig. 2.

Example

Consider detecting an alpha wave oscillation pattern:

Pattern (idealized alpha wave fragment):

- ❖ Values: `[10, 15, 12, 8, 11, 14]`
- ❖ Structure: low → peak → mid-high → low → mid → high

EEG Signal (actual recorded data):

- ❖ Window: `[45, 67, 53, 38, 49, 62]`

Despite the different amplitudes (45 vs 10, 67 vs 15), both sequences share the same relative ordering structure. OPM recognizes these as matching patterns because the relationships between consecutive elements are preserved [7], i.e., both show the same oscillation structure [6].

```
KMP-ORDER-MATCHER(T, P)
n ← |T|, m ← |P|
μ(P) ← COMPUTE-PREFIX-REP(P)
π ← KMP-COMPUTE-FAILURE-FUNCTION(P, μ(P))
𝒯 ← ϕ
q ← 0
for i ← 1 to n
    OS-INSERT(𝒯, T, i)
    r ← OS-RANK(𝒯, T[i])
    while q > 0 and r ≠ μ(P)[q + 1]
        OS-DELETE(𝒯, T[i − q..i − π[q] − 1])
        q ← π[q]
        r ← OS-RANK(𝒯, T[i])
    q ← q + 1
    if q = m
        print "pattern occurs at position" i
        OS-DELETE(𝒯, T[i − q..i − π[q] − 1])
        q ← π[q]
```

Fig. 2: KMP-Order-Matcher pseudocode for efficient OPM. T: EEG Signal; P: Pattern template.

Application to Age-Based Brain Development

In this project, OPM helps identify and track brain activity patterns across different age groups. A theta wave pattern in children might have amplitudes of [20, 35, 28, 15], while the same functional pattern in adults appears as [12, 21, 17, 9]. OPM recognizes these as representing the same cognitive state despite the amplitude differences. This enables tracking of how specific patterns appear, disappear, or change frequency with age, revealing developmental trends in brain activity.

The approach is particularly powerful when combined with Cartesian Tree Matching, as mentioned in the project description. While OPM detects patterns based on relative ordering, Cartesian Tree Matching captures hierarchical relationships and nested structures. Together, these methods provide comprehensive pattern detection that captures both sequential and structural features of brain activity, allowing for robust identification of age-dependent patterns in normal brain development.

OPM's focus on structural relationships rather than absolute values makes it robust to noise and individual variations. This ensures that the detected patterns reflect genuine neurological activity rather than measurement artifacts or individual physiological differences. By identifying which patterns occur consistently across subjects within an age group but differ between age groups, the method contributes to understanding how brain activity evolves as part of normal neurodevelopment.

2.2.2 Cartesian Tree Matching (CTM) in EEG Analysis

CTM is a pattern matching technique that detects recurring structures in EEG signals by focusing on structural properties and hierarchical relationships rather than exact values. By constructing a Cartesian Tree for each EEG segment, we represent both the order and hierarchy of signal components, independent of their amplitude, as shown in Fig.3.

Two sequences match if they have identical Cartesian Tree structures, allowing comparison of structural patterns rather than raw values.

A Cartesian Tree is a binary tree derived from a sequence that maintains two key properties: it preserves the in-order sequence (left-to-right order) and satisfies the min-heap property (parent values are smaller than their children) [5]. This captures both temporal ordering and hierarchical relationships within the data.

Example

Consider an EEG segment: `[9, 3, 7, 1, 8, 12, 10]`

Construction process:

1. Find minimum value (1) → becomes root
2. Elements to the left `[9, 3, 7]` → left subtree
3. Elements to the right `[8, 12, 10]` → right subtree
4. Recursively apply to each subtree

```
      1
     / \
    3   8
   / \   \
  9   7   10
           \
           12
```

This tree structure captures the hierarchical relationships between peaks and valleys in the EEG signal.

Pattern :

Segment A: [15, 8, 12, 5, 18, 22, 16]

Test Signal:

Segment B: [45, 24, 36, 15, 54, 66, 48]

Despite amplitude differences (Segment B values are ~3× larger), both produce identical Cartesian Tree structures. The minimum value (5 in A, 15 in B) occurs at the same position (4th), with the same hierarchical organization of surrounding peaks and valleys. This indicates that both subjects exhibit the same structural brain activity pattern during the attention task.

Application to Age-Based Brain Development

In this project, CTM identifies and tracks structural patterns across age groups. A theta-alpha coupling pattern in children might have different amplitudes than in adults, but maintain the same hierarchical organization. By quantifying how often specific structural patterns occur at each developmental stage, we can identify patterns that appear, disappear, or change frequency with age.

Complementary Analysis with OPM

CTM and OPM work together to provide a comprehensive analysis:

- ❖ **OPM**: Detects patterns based on the relative ordering of adjacent elements (local sequential relationships)
- ❖ **CTM**: Captures hierarchical organization and nested relationships (global structural properties)

While OPM identifies that values increase and then decrease, CTM reveals whether those changes form symmetric nested structures or asymmetric cascades. This combination ensures robust detection of age-dependent patterns, capturing both fine-grained temporal dynamics and broader organizational principles of brain activity.

Implementation Approach

The analysis follows these steps: segment EEG recordings into fixed-length windows, construct Cartesian Trees for each segment, compare tree structures to identify matches, and analyze pattern occurrence frequencies across age groups. This reveals which structural patterns characterize different developmental stages and how pattern distributions shift across normal brain maturation, contributing to an understanding of the organizational principles underlying cognitive development.

Fig. 3: Example pattern S = (6, 2, 5, 1, 4, 3, 7) and its corresponding Cartesian tree.

2.2.3 Motif discovery

The process of finding short, recurring shapes in EEG windows that appear many times and therefore represent “typical” neural patterns. We will first split the cleaned EEG into fixed windows (per band/channel), then encode each window’s shape (with OPM ranks or CTM parent-distance). Windows with the same or very similar codes will be grouped; groups that occur often enough (above a support threshold) become *motifs*. These motifs act like pattern “templates” for each age group: later, when we scan a new EEG, we will just compare its window codes to the motif library instead of searching blindly. This makes matching faster, more robust to noise (because we match shape, not exact amplitudes), and comparable across subjects.

3. Dataset Description

The Healthy Brain Network EEG (HBN-EEG) dataset is a large-scale collection of high-density electroencephalographic recordings from the Child Mind Institute's Healthy Brain Network initiative. The dataset includes recordings from over 3,000 children and adolescents aged 5–21 years [1], accompanied by rich psychopathology assessments and multimodal brain imaging data.

High-density EEG data were recorded at 500 Hz with a 0.1–100 Hz bandpass using a 128-channel EGI Geodesic Hydrogel system [1]. The high channel count provides excellent spatial resolution for capturing neural activity across the entire scalp. Recording sessions lasted approximately 60 minutes, during which participants performed multiple cognitive tasks following standardized protocols.

For this project, we will utilize the HBN-EEG dataset to investigate age-dependent patterns in brain activity within the 5–21 age range, to characterize normal brain development during adolescence and early adulthood. The dataset is organized according to the Brain Imaging Data Structure (BIDS) format and includes comprehensive event annotations using

Hierarchical Event Descriptors (HED), making it particularly suitable for cross-task analysis and machine learning applications.

The dataset is publicly available through the NEMAR data repository at:
https://nemar.org/dataexplorer/detail?dataset_id=ds005505
The total size of the dataset is approximately 90 GB [10], and it will be downloaded locally for analysis as part of this project.

3.1 Experimental Protocol

The HBN-EEG dataset includes EEG recordings from participants performing six distinct cognitive tasks, categorized into passive and active tasks based on the presence of user input and interaction[1]:

Passive Tasks (no motor responses required):

1. **Resting State (RS):** Participants rested with their heads on a chin rest, following instructions to open or close their eyes and fixate on a central cross. During this 5-minute task, subjects viewed a fixation cross and were prompted to open or close their eyes at different times, consisting of alternating eyes-open and eyes-closed periods.
2. **Surround Suppression (SuS):** Participants viewed flashing peripheral disks with contrasting backgrounds, while event markers and conditions were recorded. Four flashing peripheral disks were presented with a contrasting background in two runs, each approximately 3.6 minutes long.
3. **Movie Watching (MW):** Participants watched four short movies with different themes, with event markers recording the start and stop times of presentations. The movies included clips from "Despicable Me," "Fun with Fractals," "Diary of a Wimpy Kid," and "The Present."

Active Tasks (with participant behavioral input):

4. **Contrast Change Detection (CCD):** Participants identified flickering disks with dominant contrast changes and received feedback based on their responses.
5. **Sequence Learning (SL):** Participants memorized and repeated sequences of flashed circles on the screen, designed for different age groups.
6. **Symbol Search (SyS):** Participants performed a computerized symbol search task, identifying target symbols from rows of search symbols.

For our project focusing on age-related brain development patterns, we will primarily utilize the Resting State and passive viewing tasks, as these provide consistent baseline measures of brain activity that are less influenced by individual differences in task performance.

3.2 Data Format and Structure

The HBN-EEG dataset is organized according to the Brain Imaging Data Structure (BIDS) format, which provides a standardized structure for data file names, formats, and metadata fields essential for reproducibility and analysis. The dataset is available through multiple releases on platforms including NEMAR (nemar.org) and OpenNeuro.

Each release contains:

1. **EEG Data:** High-resolution recordings capturing neural activity during various tasks
2. **Behavioral Responses:** Participant responses during EEG tasks, including reaction times and accuracy, were included within the events.tsv files
3. **Hierarchical Event Descriptors (HED):** Events have clear annotations suitable for systematic meta and mega analysis
4. **Participant Information:** Demographic data (age, sex, handedness) and psychopathology dimensions

The first 11 releases of the HBN-EEG dataset are freely available under the CC-BY-SA-4.0 license. Data quality checks were performed to ensure data integrity, including checks for data length, sampling frequency, and event marker availability.

4. Related Studies Using the HBN-EEG Dataset

The following section presents a review of articles that provide insights into methodologies and findings relevant to EEG analysis, particularly those utilizing the **Healthy Brain Network EEG (HBN-EEG) dataset** [1].

These studies offer valuable perspectives on EEG data normalization strategies, pattern recognition techniques, and brain activity analysis that are directly applicable to our project of understanding normal brain development through EEG signals. By using the same dataset, these studies share common ground with our research, helping to establish a framework for identifying age-dependent brain activity patterns.

1. **Data Normalization Strategies for EEG Deep Learning**
   This study examines various normalization strategies for EEG data, with a focus on deep learning [2] models. By analyzing the **HBN-EEG** dataset, the authors explore the impact of different normalization techniques on model accuracy and performance. They emphasize the importance of proper normalization to account for variability across subjects and tasks, which directly applies to our approach of preprocessing EEG signals for pattern detection. The findings inform our project by highlighting how normalization can improve the

consistency of age-group comparisons in EEG analysis.

2. **EEG Foundation Challenge: From Cross-Task to Cross-Subject Learning**
   This paper discusses the challenges of developing generalized models for EEG data analysis across various tasks and individuals, using the **HBN-EEG** [1] dataset. The authors focus on decoding cognitive and psychopathological traits, addressing the variability inherent in EEG signals. This aligns with our project, as we are also working with multi-age group data to identify recurring EEG patterns. The study's insights on cross-subject learning will guide our efforts to detect consistent brain activity patterns across different age groups, despite individual variability.

3. **Hierarchical Deep Neural Network for Child Brain Health Assessment**
   This research presents a deep learning model for predicting brain health and mental states based on EEG signals, utilizing the **HBN-EEG** dataset. The study employs hierarchical models to classify brain activity associated with various cognitive and psychological conditions[4]. Although their focus is on identifying mental health patterns, the techniques used to analyze the **HBN-EEG** dataset apply to our project. We can leverage these deep learning approaches to better understand the normal developmental trajectories of brain activity in healthy children and adolescents.

5. Our Proposed Solution

Our solution utilizes the HBN-EEG dataset, consisting of 128-channel resting-state recordings from approximately 2,600 participants [1]. We process the raw signals through 0.5–45 Hz filtering, ICA artifact removal, and Z-score normalization to ensure quality. The participants are then stratified into Childhood (5–12 years) and Adolescence (13–21 years) cohorts. Using a time-window approach (0.5–2s), we apply Cartesian Tree Matching (CTM)[5] and Order-Preserving Matching (OPM) to detect structural features[8].

Finally, the pipeline uses motif discovery to identify distinct neural patterns specific to each age group

5.1 Research Hypothesis

We hypothesize that identifiable and recurring structural patterns may be present in EEG signals that could characterize different age groups, and that these patterns change systematically during brain maturation

Primary Hypothesis: Children (ages 5–12) may exhibit EEG patterns that differ from adolescents (ages 13–21), with differences expected to manifest as:

1. Higher dominance of Theta waves in the younger group
2. Increased Alpha wave power in the older group
3. Changes in the structural complexity of patterns

Secondary Hypothesis: OPM and CTM may reveal age-group–specific recurring patterns (motifs); their usefulness will be evaluated empirically by prevalence and generalization on held-out data.

5.2 Data Processing

Processing and cleaning EEG data is critical to ensure accurate analysis, as these signals are highly sensitive to noise and external interference. To improve signal quality, we will use various techniques such as **high-pass filters** to remove low-frequency noise like head movements or electrical oscillations, and **low-pass filters** to eliminate high-frequency noise from equipment or muscle movements [2]. Additionally, we will apply **Independent Component Analysis (ICA)** to separate and remove artifacts like eye blinks and muscle activity, isolating the brain's true electrical activity. We also apply **normalization techniques**, including subtracting the mean and dividing by the standard deviation, to ensure comparability across participants. These preprocessing steps ensure that the EEG data is clean, standardized, and suitable for identifying patterns of brain activity across different age groups.

5.3 Dataset Split

For our research, we focus on participants aged 5-21 years, dividing them into two distinct age groups to enable comparison of EEG patterns across developmental stages:

Group 1 (Early Adolescence): Ages 5-12 years
Group 2 (Late Adolescence/Young Adulthood): Ages 13-21 years

This division allows us to investigate developmental changes in brain activity patterns during the transition from early to late adolescence and into young adulthood.

Group	Age Range	Sample size	Neurological Justification
childhood	5-12 years	~1,300 participants	Theta wave dominance, high neural plasticity, and ongoing synaptic development
adolescence	13-21 years	~1,300 participants	Transition to Alpha dominance, synaptic pruning, and prefrontal cortex maturation

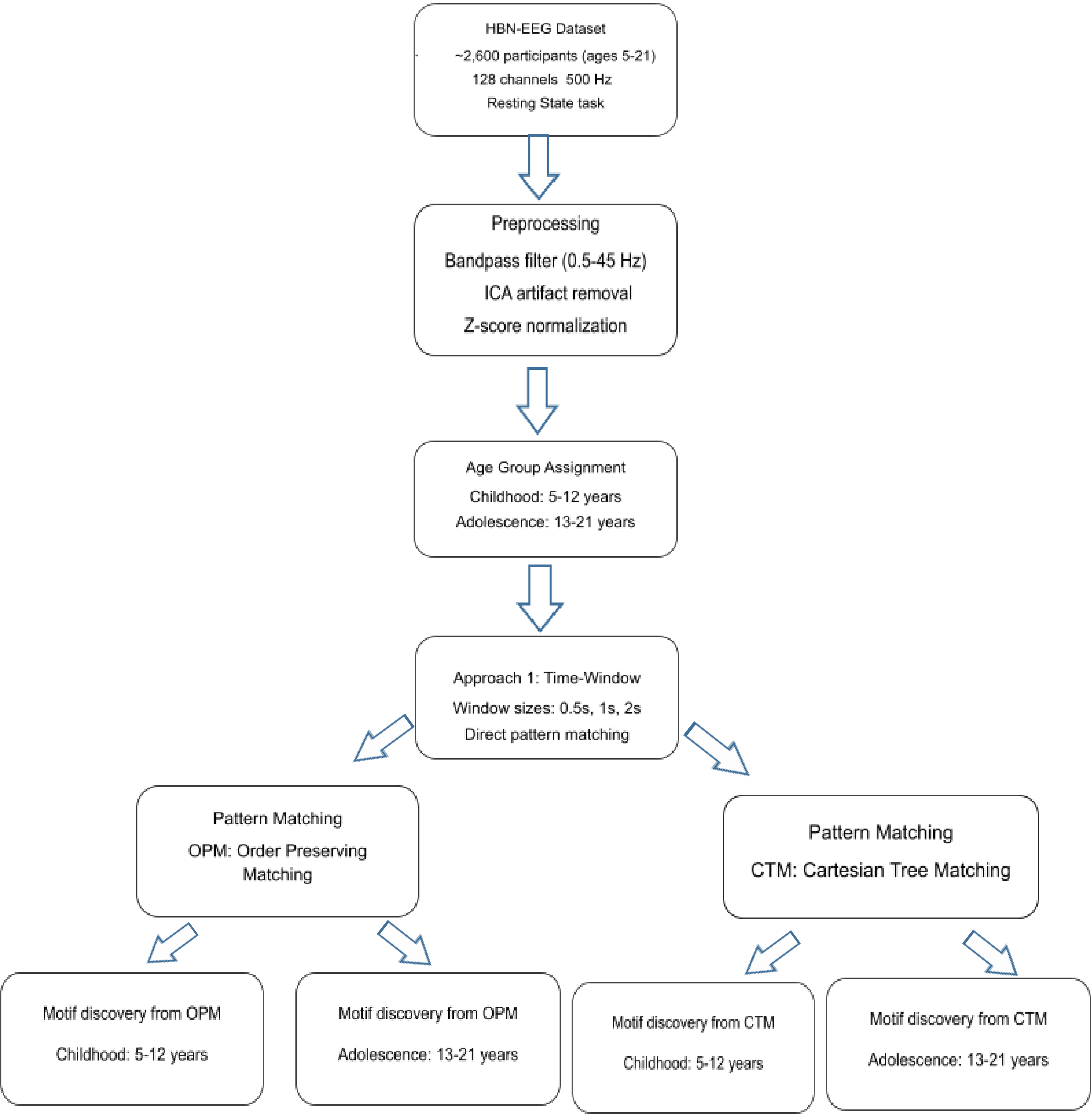
5.4 Time-Window–Based Analysis

We will split each channel’s cleaned EEG into short, overlapping windows. For every window, we will compute two codes: an OPM rank-order code and a CTM parent-distance code. These codes will drive motif discovery within age groups and matching at evaluation; per subject, we will combine scores across channels to form an age-group similarity profile.

5.5 Expected Challenges

This project is expected to face several methodological and practical challenges: EEG signals are inherently noisy and may still contain residual artifacts after preprocessing, which can affect motif stability; substantial inter-subject variability within the same age group can make it difficult to identify patterns that are both frequent and representative; the use of discrete age-group splits for a continuous developmental process may introduce boundary effects for participants near group transitions; the discovered motifs may be sensitive to design choices such as window length, overlap, frequency-band selection, and support thresholds; applying OPM/CTM across many channels and windows may create notable computational costs; unequal sample distribution across ages may bias motif libraries toward more represented ranges; and even when recurring motifs are detected, interpreting them in a biologically meaningful developmental context remains nontrivial. In addition, conclusions derived from HBN-EEG may not fully generalize to datasets collected under different protocols, so careful sensitivity analysis, transparent parameter reporting, and explicit limitation statements are necessary to ensure robust and credible findings.

6. Process Flow Chart



7.Verification plan

Case No.	Step	Action Description	Expected Result	Success Criteria
1	Data Loading	Load HBN-EEG dataset in BIDS format	All recordings loaded correctly	No loading errors, correct channel count ((128
2	Preprocessing	Band-pass 0.5–45 Hz (+ optional 50/60 Hz notch) → ICA artifact removal (blinks/EMG)	Clean broadband signal per channel	PSD outside 0.5–45 Hz < 1%; visual check: artifacts removed
3	Normalization	Z-score per participant × channel	Comparable scale across subjects	
4	Age split	Assign groups: 5–12 vs 13–21	Two cohorts prepared	Group sized within ±10%
5	Windowing	Fixed windows (e.g., L = 0.5/1/2 s; hop 50–100%) per channel	Window set covering recordings	time 95% < coverage; no gaps
6	Encoding	For each window: OPM rank code + CTM parent-distance code	Codes stored with (subject, age, chan, t0)	Code lengths valid; no encoding errors
7	Motif discovery	Per (age, channel): cluster frequent)codes (OPM/CTM	Motif library per group/channel	K motifs found with ≥ support ≥ τ

8.AI Tools Usage

We used AI tools to support different stages of our research and development process. ChatGPT was used to brainstorm ideas, clarify technical concepts, and help refine our project scope and methodology. It also supported writing tasks such as improving phrasing, organizing sections, and producing structured summaries. Google Gemini was used to quickly review and summarize research materials, compare approaches across papers, and help extract key points that informed our design decisions. Using these AI tools improved our productivity and helped us build a clearer, more organized plan for implementing our proposed solution.

Example prompt used in ChatGPT:

Explain Order-Preserving Matching (OPM) in simple terms and give an EEG-based example.

Example prompt used in Gemini:

Extract the main normalization strategies discussed in this paper and explain when each is useful.

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